

Scientific Vita

(Wissenschaftlicher Lebenslauf)

Christoph Gerhardt

born July 17, 1990 in Ilmenau, Germany

Education

- Oct 2019 – June 2026 (expected) Dr.-Ing. (PhD) in Computer Science**
Technische Universität Ilmenau, Faculty of Computer Science and Automation
Dissertation: "Comprehensive Outdoor Scene Understanding Through Multi-Scale Neural Networks: From 360° Object Detection to Panoramic Segmentation and Weather-Robust Augmentation"
Supervisor: Prof. Dr. rer. nat. Wolfgang Broll
Research Group: Virtual Worlds and Digital Games (VWDG)
- Oct 2014 – Sep 2017 M.Sc. Computer Engineering (Ingenieurinformatik)**
Technische Universität Ilmenau — Final Grade: 1.4 (very good)
Specialisation: Multimedia Information and Communication Systems.
Thesis: "Selektive Verschlüsselung von Gesichtern in Videodaten unter Verwendung des H.264-Standards"
- Oct 2010 – Sep 2014 B.Sc. Computer Engineering (Ingenieurinformatik)**
Technische Universität Ilmenau — Final Grade: 1.8 (good)
Thesis: "Entwicklung eines Verfahrens zur Verbesserung von Mikrofonklassifizierung basierend auf der Analyse von Umwelteinflüssen"
- Jun 2009 Abitur**
Gymnasium "Am Lindenberg", Ilmenau — Grade: 1.5

Professional Experience

- Feb 2019 – present Research Associate (Wissenschaftlicher Mitarbeiter)**
Technische Universität Ilmenau, Fachgebiet Virtuelle Welten und Digitale Spiele
Doctoral research on outdoor scene understanding with neural networks; development of datasets, architectures, and benchmarks; teaching; student supervision

- Feb 2017 – Jan 2019 Software Developer**
Fraunhofer Institute for Digital Media Technology (IDMT), Ilmenau
Group: Virtual Acoustics — C++ and Python development for spatial audio rendering
- Jul 2016 – Jan 2017 Student Research Assistant (Hilfswiss. Mitarbeiter)**
Fraunhofer IDMT, Ilmenau
Group: Media Distribution and Security — Face detection and selective encryption in H.264 video

Research Activities

Dissertation Research (2019–2026)

My doctoral research addresses the challenge of **comprehensive outdoor scene understanding** through a progression of increasingly complex perception tasks:

1. **360° Traffic Sign Detection** — Adapted object detection networks for equirectangular panoramic images; created a dataset of 2 500 images with 81 traffic sign classes; deployed in NOVASIB GmbH's TT-SIB INFOSYS® road inventory system (IEEE ITSC 2020).
2. **Universal Outdoor Scene Segmentation (OUTSIDE)** — Designed a multi-scale semantic segmentation architecture for unconstrained outdoor scenes; curated the OUTSIDE15k dataset (15 000 images, 24 classes) (IEEE MMSP 2021).
3. **Sky and Cloud Segmentation (SkyCloud)** — Developed SkyCloudNet for fine-grained sky and cloud segmentation from perspective images; released dataset and pre-trained models (IEEE ICIVC 2023).
4. **Panoramic Sky Segmentation (SkyCloud360)** — Extended sky/cloud segmentation to equirectangular 360° imagery; created a 600-image panoramic dataset with addressing of projection distortions (IEEE ICIVC 2025).
5. **Weather-Robust Augmentation (AWARE)** — Developed a three-pipeline benchmark framework (SWIFT, PRISM, PROVE) evaluating 25 augmentation strategies for weather robustness; generated the AWACS dataset (71 400 images) (IEEE Access, 2026, 10.1109/ACCESS.2026.3699854).

Collaborative Research

In addition to the core dissertation work, I contributed to collaborative research within the VWDG group on:

- **Avatar and Agent Realism** — Behavioural realism of animated avatars in AR and VR environments, including single-point IK animation and full-body motion capture comparisons (IEEE VR 2024, Int. Conf. Cyberworlds 2024, The Visual Computer 2025).
- **AR Communication Systems** — User studies with older adults using wearable AR communication systems (ACM IMX 2024, ITAP/HCI 2025).
- **Gaze and Audiovisual Coherence** — Analysis of gaze behaviour and effects of visual realism and audio presentation in AR storytelling (ACM ETRA 2024, IEEE VR 2024).
- **Systematic Reviews** — Co-authored a systematic review on avatar and agent visualisation in HMD-based AR & VR (IEEE TVCG 2023).
- **Telepresence Quality** — Impact of partner representation (robot, avatar, human) on communication experience (ACM CSCW 2025).

Open-Source Contributions & Datasets

- **AWARE Framework:** SWIFT + PRISM + PROVE pipelines (<https://carhartt21.github.io/AWARE>)
- **AWACS Dataset:** 71 400 weather-augmented images
- **SkyCloud360:** 600 equirectangular images (<https://github.com/carhartt21/SkyCloud360>)
- **SkyCloudNet:** Sky/cloud segmentation models (<https://github.com/carhartt21/SkyCloudNet>)
- **OUTSIDE15k:** 15 000 images, 24 classes (<https://github.com/carhartt21/outsideNet>)
- **360° Detection:** 2 500 panoramic images, 81 sign classes (<https://github.com/carhartt21/360-Object-Detection>)

Conference Presentations

1. **Oral Presentation:** “SkyCloud360: Sky and Cloud Segmentation in Equirectangular Images,” 10th Int. Conf. Image, Vision and Computing (ICIVC), Chengdu, China, June 2025.
2. **Oral Presentation:** “SkyCloud: Neural Network-Based Sky and Cloud Segmentation from Natural Images,” 8th Int. Conf. Image, Vision and Computing (ICIVC), Dalian, China, August 2023.
3. **Oral Presentation:** “OUTSIDE: Multi-Scale Semantic Segmentation of Universal Outdoor Scenes,” IEEE 23rd Int. Workshop Multimedia Signal Processing (MMSP), Tampere, Finland, October 2021.
4. **Oral Presentation:** “Neural Network-Based Traffic Sign Recognition in 360° Images for Semi-Automatic Road Maintenance Inventory,” IEEE 23rd Int. Conf. Intelligent

Transportation Systems (ITSC), virtual, September 2020.

Teaching Activities

Courses (Teaching Assistant)

SS 2026	Virtual and Augmented Reality Technische Universität Ilmenau
WS 2025/26	Game Development Technische Universität Ilmenau

Master Thesis Supervision

- **Zamam Mirza:** "Synthetic Data Generation for Semantic Segmentation and Object Detection Using Generative Models"
- **Syed Ajaz Zaidi:** "Enhancing Digital Reading with Gaze-Triggered Audio Cues and Audiobook Listening with Augmented Sound Effects"

Research Project Supervision

SS 2024	Generative Models for Data Augmentation
WS 2024/25	Mimic Synthesis for 3D Avatars using Gaze and Audio; Automatic Non-Reference Quality Assessment for Artificially Generated Images; Cross-Domain Image-to-Image Translation for Low-Light Image Enhancement; 3D Avatar Generation from Monocular Video; Guided Image Generation Using Stable Diffusion Models; Realistic Hair Generation for 3D Avatars using Deep Learning
SS 2025	Data Augmentation in Adverse Outdoor Conditions using Simulator Data; Utilizing Simulator-Generated Data for Enhanced Model Training; Non-Reference Quality Assessment for Synthetic Images
WS 2025/26	Non-Reference Quality Assessment for Generated Images; Large Tutoring Models; Egocentric Video Pretraining for XR Assistants; LLM-Driven Facial Animations for Avatars
SS 2026	Image-to-Image Translation in Equirectangular/360° Images; Synthetic Fog Augmentation for Semantic Segmentation; Unsupervised Domain Adaptation for Adverse Weather Conditions; Benchmarking Vision-Language Models for Adverse Weather Scene Understanding

Reviewer Activities

Journal Reviewing

- **IEEE Transactions on Visualization and Computer Graphics (TVCG)**
“Towards Privacy-preserving Photorealistic Self-avatars in Mixed Reality”
- **IEEE Access**
“Embodiment and Presence in Virtual Reality: The Role of Self-Similarity in High-Fidelity 3D Avatars”
- **IEEE Open Journal of Signal Processing**
“Unsupervised Angularly Consistent 4D Light Field Segmentation using Hyperpixels and a Graph Neural Network”

Conference Reviewing

- **IEEE ISMAR 2025**
“Taking Language Embedded 3D Gaussian Splatting into the Wild”
- **IEEE ISMAR 2024**
“Exploring Device-Oriented Video Encryption for Hierarchical Privacy Protection in AR Content Sharing”
“HOIMotion: Forecasting Human Motion During Human-Object Interactions Using Egocentric 3D Object Bounding Boxes”
“Comparative Analysis of AR-Based and Traditional Approaches in English Pronunciation Teaching: Efficacy, Engagement, and Implementation”
- **ACM Multimedia 2026**
“CoSpeakAR: Proactive In-situ Scaffolding for L2 Spoken Communication via AR”
“A Mobile Augmented Reality System for Persistent Virtual Human Interaction”
“MONET: Museum Observatory of Neuro-affective Eye-Tracking”
“A Natural Confidant: Exploring the Impact of a Virtual Natural Agent on Emotional Responses and Immersion in Conversational VR”

Ilmenau, June 5, 2026